

Variables

Learning Objectives

- Create and use variables in Scratch
- Understand variables as containers for storing values
- Use variables to track scores and counts

Engage (5-7 minutes)

Think of a donation box at a temple. You put money in, take it out, and the amount changes. Variables in Scratch work like donation boxes—they store numbers or text that can change while your program runs. Today we learn to use these useful containers.

Explore (10-12 minutes)

Click on Variables and then “Make a Variable.” Name it “score” and watch it appear on the stage. Use “set score to 0” and “change score by 1” blocks. Create a project where clicking a sprite increases the score. Display the variable on stage and watch it update live.

Explain (10-12 minutes)

Variables are named containers that store data. You create them, give them a starting value, and change that value as your program runs. “Set” gives a variable a fixed value, while “change” adds to or subtracts from it. Variables help you keep track of scores, counts, and other changing information.

Elaborate (8-10 minutes)

Build a simple quiz game using variables. Create a “score” variable that increases with correct answers and a “lives” variable that decreases with wrong ones. Show both on the stage. Like a village cricket match scoreboard, your variables track the game progress in real time.

Evaluate (5-8 minutes)

Exit ticket: What is a variable? If you set “score” to 10 and then use “change score by 3,” what is the new value? Name one real-life example of something that works like a variable.